

QMC and Importance methods

Lecturer: Riccardo Corradin

Quasi Monte Carlo

—

From MC to QMC

Many problems require evaluating an integral of a specific function, say $f(\mathbf{x})$, $\mathbf{x} \in \mathbb{X} \subseteq \mathbb{R}^d$.

→ What we see regarding QMC do not strictly relate to Bayesian inference, but computing integrals in general.

With standard **Monte Carlo integration** theory, we want to compute

$$\mu_g = \mathbb{E}_f[g(\mathbf{x})] = \int_{\mathbb{X}} g(\mathbf{x})f(\mathbf{x})d\mathbf{x},$$

where f can be either continuous or discrete.

Standard approach: we produce a sample $X_1, X_2, \dots, X_n$ where the generic $X_i \sim f$, and perform the integration via Monte Carlo as

$$\hat{\mu}_g = \frac{1}{n} \sum_{i=1}^n g(X_i).$$

- If $\mathbb{E}_f[g(x)]$ is finite, by the **strong law of large number**, $\hat{\mu}_g \xrightarrow{a.s.} \mu$, as n diverges.
- If also $\mathbb{E}_f[g(x)^2]$ is finite, by the **central limit theorem**, $(\hat{\mu}_g - \mu) \xrightarrow{d} N(0, \sigma_g^2/n)$, as n diverges, with σ_g^2 being the variance of $g(x)$ (we can estimate this term).
- The convergence rate, thanks to the **Berry–Esseen theorem** of the MC estimator is of the order $O(1/\sqrt{n})$.

From MC to QMC

First, note that with suitable change of variables, we can write the previous integration in a bounded domain $[0, 1)^d$, having

$$\mathbb{E}_{\mathbf{u}}[h(\mathbf{u})] = \int_{[0,1)^d} h(\mathbf{u}) d\mathbf{u},$$

where $\mathbf{u} = \psi(\mathbf{x})$ a suitable transformation and $h(\mathbf{u})$ corresponds to $g(F^{-1}(\mathbf{u}))$.

- Monte Carlo methods relate to approximate the integral w.r.t. **uniformly distributed random variables**.
- Sequences produced to perform QMC integration aim to **approximate such uniform distribution**.

We want to evaluate the following integral, where $h : \mathbb{R}^d \rightarrow \mathbb{R}$,

$$I = \int_{\mathbb{R}^d} \cos(\|\mathbf{x}\|) e^{-\|\mathbf{x}\|^2} d\mathbf{x}$$

with $\|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_d^2}$. We immediately note that

$$I = \int_{\mathbb{R}^d} \pi^{d/2} \cos(\|\mathbf{x}\|) \frac{e^{-\|\mathbf{x}\|^2}}{\pi^{d/2}} d\mathbf{x} = \mathbb{E}_{\mathbf{X}} \left[\pi^{d/2} \cos(\|\mathbf{x}\|) \right], \quad \mathbf{x} \sim N(\mathbf{0}, \frac{1}{2} I_d).$$

From MC to QMC

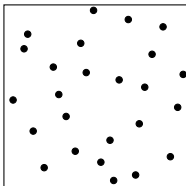
At first, the previous is equivalent to perform a change of variable with $\mathbf{x} = (\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d))/\sqrt{2}$, with Φ being the cdf of the univariate Gaussian distribution, and

$$I = \int_{\mathbb{R}^d} \pi^{d/2} \cos \left(\left\| (\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_d)) \right\| / \sqrt{2} \right) d\mathbf{u} = \mathbb{E}_U [h(\mathbf{u})] \approx \frac{1}{n} \sum_{i=1}^n h(\mathbf{U}_i),$$

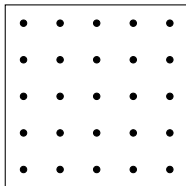
with $\mathbf{U}_i \sim \text{Unif}([0, 1]^d)$, $i = 1, \dots, n$.

The underlying idea of QMC is to replace the randomly-sampled values $\mathbf{U}_i$ with a deterministic sequence that mimics the uniform distribution. But how we choose the sequence?

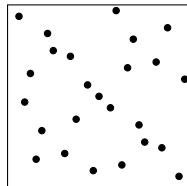
Uniform random sample



Equally spaced lattice



Sobol sequence



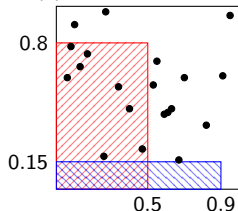
Discrepancy with QMC

[Discrepancy] Let $\mathcal{A}$ be a collection of (measurable) subsets of $[0, 1]^d$. The discrepancy of the points $\{u_1, \dots, u_n\}$ relative to $\mathcal{A}$ is given by

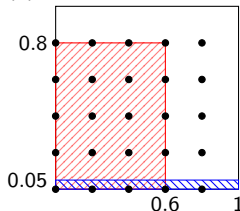
$$D_{\mathcal{A}}(\mathbf{u}_n) = \sup_{A \in \mathcal{A}} \left| \frac{\#\{\mathbf{u}_n \in A\}}{n} - \text{vol}(A) \right|.$$

- The first term is a proportion (in $[0, 1]$), the second term is the volume of a subset of $[0, 1]^d$, having total volume 1, hence $D_{\mathcal{A}}(\mathbf{u}_n) \in [0, 1]$.
- $D_{\mathcal{A}}(\mathbf{u}_n)$ measures of non-uniformity, as it gets larger when (i) many points are in a small subset $A \in \mathcal{A}$ or (ii) only few points are in a large subset $A \in \mathcal{A}$.
- The ordinary (or extreme) discrepancy corresponds to the case of $\mathcal{A}$ being the collection of all rectangles in $[0, 1]^d$ of the form $\prod_{j=1}^d [\ell_j, u_j]$, $0 \leq \ell_j \leq u_j \leq 1$. The star discrepancy consider the rectangles anchored to the origin.

(a) Some $n = 20$ points



(b) A lattice of $n = 25$ points



Koksma-Hlawka inequality

[Koksma-Hlawka inequality] If the function h has finite Hardy-Krause variation $V(h)$ on $[0, 1]^d$, then, for any $\mathbf{u}_1, \dots, \mathbf{u}_n \in [0, 1]^d$ we have

$$\left| \frac{1}{n} \sum_{i=1}^n h(\mathbf{u}_i) - \int_{[0,1]^d} h(\mathbf{u}) d\mathbf{u} \right| \leq V(h) D_{\mathcal{A}}(\mathbf{u}_n)$$

- Hardy-Krause variation is a measure of the total variation for multivariate functions. It is the sum of total variations of all the partial derivatives and boundary restrictions.
 - If h is regular enough, it is finite (e.g. $h \in C^d([0, 1]^d)$).
- **Fundamental fact for us:** the error bound for QMC (or in general for a MC procedure) depends on two term
 - The HK variation of the function, that does not depend on specific choices of the evaluation points.
 - The discrepancy measure, that does not depends on the specific function we want to integrate.but they are difficult to compute.

[Low-discrepancy sequence] A sequence of points $\mathbf{u}_1, \mathbf{u}_2, \dots \in [0, 1]^d$ is a low-discrepancy sequence if for any n we have

$$D_{\mathcal{A}}(\mathbf{u}_n) \leq C_n \frac{\log(n)^d}{n},$$

where C_d depends only on the dimension.

- Hence, a low-discrepancy sequence has $O(1/n^{1-\epsilon}) \approx O(1/n)$ order of convergence.

Van der Corput sequence

There are a lot of strategies to generate a grid of deterministic points that look like uniformly distributed. Among them, we consider the van der Corput sequence at first, to generate univariate sequences.

- Each integer number can be represented in a binary base, with

$$n = n_0 + 2n_1 + 2^2n_2 + 2^3n_3 + \dots, \quad n_i \in \{0, 1\}.$$

For example $13 = (1101)_2$ so that we have $13 = 1 + 2 \times 0 + 2^2 \times 1 + 2^3 \times 1$.

- Now take the binary digits of a number n and reflect them across the binary point

$$x_n = \frac{n_0}{2} + \frac{n_1}{2^2} + \frac{n_2}{2^3} + \frac{n_3}{2^4} + \dots$$

An example with the first 5 numbers

n	binary	x_n
0	000	0
1	001	0.5
2	010	0.25
3	011	0.75
4	100	0.125

Sobol sequence

Sobol sequence extends the previous to the multivariate case.

- For each dimension $j = 1, \dots, d$, Sobol defines $v_1^{(j)}, v_2^{(j)}, \dots$ called direction numbers, that act on how we define the sequence. These are binary fractions, for example

$$v_1 = 0.10000_2 = \frac{1}{2}, \quad v_2 = 0.01000_2 = \frac{1}{4}, \quad \dots$$

- The Sobol coordinates are then defined as

$$x_n^{(j)} = n_0 v_1^{(j)} \oplus n_1 v_2^{(j)} \oplus n_2 v_3^{(j)} \oplus \dots,$$

where $\oplus$ denotes the XOR operator and $v_i^{(j)}$ the j th dimension variation of the i th direction number.

- The XOR operator allow us to stay within $\{0, 1\}$ for each digit of the binary fractions, since

$$1 \oplus 1 = 0 \oplus 0 = 0, \quad 1 \oplus 0 = 0 \oplus 1 = 1.$$

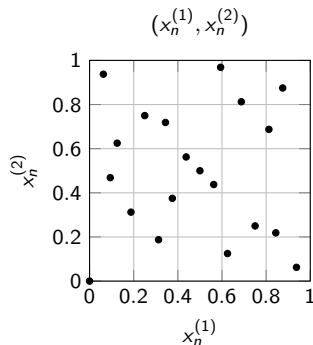
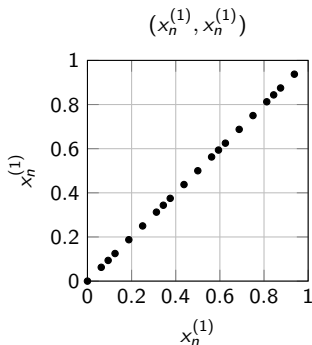
- For the first dimension, we have $v_i^{(1)} = v_i$, hence we do not change the starting sequence, and it collapses on the van der Corput sequence.
- For each other dimension, we consider primitive polynomials in $\{0, 1\}$ to induce a perturbation of the values. For example, in the second dimension we consider $P(x) = x + 1$, etc....

Sobol sequence

Practical example, $n = 4$, $d = 2$, we have

n	binary	$v_i^{(1)}$	$v_i^{(2)}$	$x_i^{(1)}$	$x_i^{(2)}$
0	000	0.10000_2	0.10000_2	0	0
1	001	0.01000_2	0.01000_2	0.5	0.5
2	010	0.00100_2	0.01100_2	0.25	0.25
3	011	0.00010_2	0.00010_2	0.75	0.75
4	100	0.00001_2	0.00011_2	0.125	0.375

Why we need perturbations with polynomials?



Randomized Sobol sequence

A common strategy is to combine a source of randomness with the previous Sobol sequence. There are many strategies to do that, one common approach is the so-called **digital shift**.

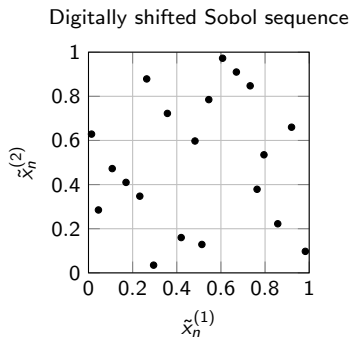
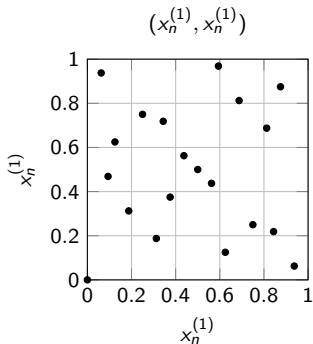
- We sample $\mathbf{u} \sim \text{Unif}([0, 1]^d)$, then we combine all the previous values as

$$\tilde{\mathbf{x}}_i = [\mathbf{x}_i]_2 \oplus [\mathbf{u}]_2,$$

i.e. we apply the XOR operator to each digit in a binary representation

Randomizing strategies can further improve the rate of convergence of our estimators

- See, e.g., Owen (1997), with rate of convergence approximately of order $O(n^{3/2})$.



Results

Back on our example integral, a quadrature gives us 2.165929 with absolute error $< 1.4 \times 10^{-11}$

Method	$n = 1$	$n = 10$	$n = 20$	$n = 30$
Lattice	9.870 (0.000)	2.714 (0.000)	2.405 (0.000)	2.312 (0.000)
Sobol	9.870 (0.000)	2.168 (0.000)	2.166 (0.000)	2.166 (0.000)
R-Sobol	1.972 (4.309)	2.166 (0.001)	2.166 (0.000)	2.166 (0.000)
Unif	2.130 (3.997)	2.171 (0.042)	2.166 (0.010)	2.167 (0.004)

Table 1: Estimated values (with SE)

Method	$n = 1$	$n = 10$	$n = 20$	$n = 30$
Lattice	0.000 (0.000)	0.016 (0.000)	0.226 (0.000)	1.404 (0.000)
Sobol	0.000 (0.000)	0.022 (0.000)	0.212 (0.000)	1.143 (0.000)
R-Sobol	0.000 (0.000)	0.020 (0.024)	0.250 (0.043)	1.243 (0.090)
Unif	0.000 (0.001)	0.014 (0.004)	0.242 (0.040)	1.214 (0.088)

Table 2: Computational times (with SE)

Importance methods

Welcome back importance sampling

- **Importance sampling** is one of the most commonly used Monte Carlo methods, which often can simplify complex problem by resorting to instrumental distributions.
- We assume the usual setting, where we want to produce a sample from the posterior distribution of $\theta \mid \mathbf{X}$, where $\theta \in \Theta \subseteq \mathbb{R}^p$ is the parameter of interest, $\mathbf{X} \in \mathbb{X}^n$ denotes the observed data, with $\mathbb{X} \in \mathbb{R}^d$.
- As usual $\pi(\theta \mid \mathbf{X}) \propto L(\mathbf{X} \mid \theta)\pi(\theta)$ the density function of the posterior distribution.
- Recall that, in a Monte Carlo setting, we want to estimate

$$\mathbb{E}_{\theta \mid \mathbf{X}} [g(\theta)] = \int_{\Theta} g(\theta) \pi(\theta \mid \mathbf{X}) d\theta,$$

by resorting to a sampling procedure

$$\frac{1}{R} \sum_{r=1}^R g(\theta^{(r)}), \quad \theta^{(r)} \stackrel{iid}{\sim} \pi(\theta \mid \mathbf{X}).$$

- Importance sampling, among other motivations, is a suitable sampling strategy when we can **easily evaluate the density function** $\theta \mid \mathbf{X}$, but it is **difficult to sample** from such a distribution.

Rejection sampler, an old friend

- We recall one of the basic Monte Carlo sampler, the rejection sampler.
- Suppose we want to sample from $\pi(\boldsymbol{\theta} \mid \mathbf{X})$ using an auxiliary distribution $h(\boldsymbol{\theta})$. With rejection sampling we can produce a sample as follows.

At the r th sampling step

- i) We generate $\boldsymbol{\theta}^{(r)} \stackrel{iid}{\sim} h(\boldsymbol{\theta})$ (independent of the previous state).
- ii) Accept $\boldsymbol{\theta}^{(r)}$ with probability

$$\frac{\pi(\boldsymbol{\theta}^{(r)} \mid \mathbf{X})}{Kh(\boldsymbol{\theta}^{(r)})}, \quad \text{with} \quad K \geq \max_{\boldsymbol{\theta}} \frac{\pi(\boldsymbol{\theta}^{(r)} \mid \mathbf{X})}{h(\boldsymbol{\theta}^{(r)})},$$

otherwise go back to i).

- Here we still have to evaluate the likelihood function, we can extend this algorithm accommodating for synthetic data.
- The support of $h(\cdot)$ should cover the support of $\pi(\boldsymbol{\theta}^{(r)} \mid \mathbf{X})$.

Normalised importance sampling

- Importance sampling get advantage of knowing another density function, say $q(\theta)$, for which $\pi(\theta | \mathbf{X}) \ll q(\theta)$.
- In practice, we use the distribution $q(\theta)$ to sample, but we also exploit its relation with the targeted distribution.
- The crucial part associated with importance sampling (IS) is summarized in the following identity

$$\int_{\Theta} g(\theta) \pi(\theta | \mathbf{X}) d\theta = \int_{\Theta} g(\theta) \frac{\pi(\theta | \mathbf{X})}{q(\theta)} q(\theta) d\theta,$$

where $w(\theta) = \frac{\pi(\theta | \mathbf{X})}{q(\theta)}$ is called the importance weight function.

- The previous identity holds if $q(\theta) > 0$ whenever $g(\theta)\pi(\theta | \mathbf{X}) > 0$.
- This justify the use of the following Monte Carlo estimator

$$\int_{\Theta} g(\theta) \pi(\theta | \mathbf{X}) d\theta \approx \frac{1}{R} \sum_{r=1}^R w(\theta^{(r)}) g(\theta^{(r)}), \quad w(\theta) = \frac{\pi(\theta | \mathbf{X})}{q(\theta)}, \quad \theta^{(r)} \stackrel{iid}{\sim} q(\theta).$$

Accuracy of IS

- For a generic unbiased Monte Carlo estimator,

$$MSE_{\pi} \left(\frac{1}{R} \sum_{r=1}^R g(\boldsymbol{\theta}^{(r)}) \right) = \frac{1}{R} \text{Var}_{\pi}(g(\boldsymbol{\theta})).$$

- The IS estimator is unbiased, since

$$\mathbb{E}_q \left[\frac{1}{R} \sum_{r=1}^R w(\boldsymbol{\theta}^{(r)}) g(\boldsymbol{\theta}^{(r)}) \right] = \mathbb{E}_{\pi}[g(\boldsymbol{\theta})].$$

- The MSE committed while doing IS is equal to

$$MSE \left(\frac{1}{R} \sum_{r=1}^R w(\boldsymbol{\theta}^{(r)}) g(\boldsymbol{\theta}^{(r)}) \right) = \frac{1}{R} \text{Var}_q(w(\boldsymbol{\theta})g(\boldsymbol{\theta})).$$

- The IS estimator is much useless if $\mathbb{E}_q[w(\boldsymbol{\theta})^2 g(\boldsymbol{\theta})^2] < \infty$ does not hold.
- A sufficient condition is that $w(\boldsymbol{\theta})$ is upper bounded in $\boldsymbol{\theta}$.

- Among all the possible importance distribution, we can find an optimal one which maximize the accuracy of the sampler.

Theorem

The MSE of the normalized IS estimator of $E_{\pi}[g(\theta)]$ is minimized by

$$\tilde{q}(\theta) = \frac{\pi(\theta \mid \mathbf{X})|g(\theta)|}{\int_{\Theta} |g(\theta)| \pi(\theta \mid \mathbf{X}) d\theta}.$$

- Unfortunately, the previous is quite useless given that we need to compute the denominator, which contains a similar functional to the one we try to avoid.
- In practice, we want the proposal distribution to be close as possible to $\pi(\theta \mid \mathbf{X})$ within a class of tractable distribution.
- Further, we can tune the proposal distribution depending on specific functional we are considering.

IS in practice

- One of the main application of IS is when we are interested on specific functional of the posterior distribution.
- For example, if we want to integrate distributional tails with extreme quantiles, associated to rare event probabilities.

Suppose we want to quantify the probability mass that a standard Gaussian distribution $\pi(\theta | \mathbf{X})$ is leaving on the right tail, after the value $\theta = 10$, i.e. $1 - \Phi(10)$. Hence, the functional of interest is given by $g(\theta) = \phi(\theta)1_{[\theta > 10]}$. Resorting to other samplers, the accuracy on the tail is quite low since is rare to sample values on that part of the support.

With a generic IS the accuracy can also be quite low. A more smart proposal distribution concentrate the effort in the relevant part of the support. Hence, we can consider an IS with

$$q(\theta) = \lambda e^{-\lambda(\theta - c)} 1_{[\theta \geq c]},$$

where all the sampling is concentrated on relevant part of the support, and then weighted by

$$w(\theta) = \frac{\pi(\theta | \mathbf{X})}{q(\theta)} = \frac{1}{\lambda\sqrt{2\pi}} \frac{e^{-\theta^2/2}}{e^{-\lambda(\theta-10)}} 1_{[\theta \geq 10]},$$

the ratio between the density function of a gaussian distribution and a shifted exponential distribution. You can show that the optima choice leads to $\lambda = 10$, matching the hazard rate of Gaussian tails.

Auto-normalised importance sampling

- In practice, the posterior distribution of interest is usually known up to a normalization constant. But usually, we can easily evaluate

$$\pi(\boldsymbol{\theta}, \mathbf{X}) = L(\mathbf{X} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta}).$$

- Similarly, let us consider $q(\boldsymbol{\theta}) = \tilde{q}(\boldsymbol{\theta})/C_q$, with C_q being the normalization constant of $\tilde{q}(\boldsymbol{\theta})$.
- The IS works also up to normalization constants of both target distribution and proposal distribution, since

$$\int_{\Theta} g(\boldsymbol{\theta})\pi(\boldsymbol{\theta} \mid \mathbf{X})d\boldsymbol{\theta} = \frac{\int_{\Theta} g(\boldsymbol{\theta}) \frac{\pi(\boldsymbol{\theta} \mid \mathbf{X})}{q(\boldsymbol{\theta})} q(\boldsymbol{\theta})d\boldsymbol{\theta}}{\int_{\Theta} \frac{\pi(\boldsymbol{\theta} \mid \mathbf{X})}{q(\boldsymbol{\theta})} q(\boldsymbol{\theta})d\boldsymbol{\theta}} = \frac{\int_{\Theta} g(\boldsymbol{\theta}) \frac{\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{X})}{\tilde{q}(\boldsymbol{\theta})} q(\boldsymbol{\theta})d\boldsymbol{\theta}}{\int_{\Theta} \frac{\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{X})}{\tilde{q}(\boldsymbol{\theta})} q(\boldsymbol{\theta})d\boldsymbol{\theta}},$$

which leads to the following Monte Carlo estimator

$$\mathbb{E}_{\pi}[g(\boldsymbol{\theta})] \approx \frac{\sum_{r=1}^R w(\boldsymbol{\theta}^{(r)})g(\boldsymbol{\theta}^{(r)})}{\sum_{r=1}^R w(\boldsymbol{\theta}^{(r)})}, \quad w(\boldsymbol{\theta}) = \frac{\tilde{\pi}(\boldsymbol{\theta} \mid \mathbf{X})}{q(\boldsymbol{\theta})}, \quad \boldsymbol{\theta}^{(r)} \stackrel{iid}{\sim} q.$$

- Auto-normalized IS is a **biased** estimator, nevertheless
 - It is consistent, so is asymptotically unbiased
 - The bias is of an order of $O(1/\sqrt{R})$
 - With suitable choices of q , the MSE can be lower than unbiased estimators.

A formal viewpoint of IS

- Instead of thinking as density function $\pi(\boldsymbol{\theta} \mid \mathbf{X})$ and $q(\boldsymbol{\theta})$, we can view the IS in terms of probability measure $\mathbb{P}$ and $\mathbb{Q}$.
- We assume $\mathbb{P} \ll \mathbb{Q}$.
- For any test function $g(\cdot)$, we denote the expectation w.r.t. $\mathbb{P}$ as

$$\mathbb{P}(g) := \int_{\Theta} g(\boldsymbol{\theta}) \mathbb{Q}(d\boldsymbol{\theta}).$$

- Recall that the (**potentially un-normalized**) Radon-Nikodym derivative between $\mathbb{P}$ and $\mathbb{Q}$ is given by

$$w(\boldsymbol{\theta}) \propto \frac{\mathbb{P}(d\boldsymbol{\theta})}{\mathbb{Q}(d\boldsymbol{\theta})},$$

meaning that

$$\mathbb{P}(A) = \int_A w(\boldsymbol{\theta}) \mathbb{Q}(d\boldsymbol{\theta}), \quad \forall A \subseteq \Theta.$$

Theorem

Let $w(\boldsymbol{\theta}) \propto \mathbb{P}(d\boldsymbol{\theta})/\mathbb{Q}(\boldsymbol{\theta})$. Then, for any test function g , we have

$$\mathbb{Q}(gw) = \mathbb{P}(g)\mathbb{Q}(w).$$

A formal viewpoint of IS

- In force of the previous identity, we can rethink at the IS as a change of measure, where we are changing from $\mathbb{Q}$ to $\mathbb{P}$.
- From the previous identity,

$$\mathbb{Q}(gw) = \mathbb{P}(g)\mathbb{Q}(w),$$

we have the following estimator of $\mathbb{P}(g)$

$$\mathbb{P}(g) \approx \sum_{r=1}^R W^{(r)} g(\theta_r), \quad W^{(r)} = \frac{w(\theta_r)}{\sum_{r=1}^R w(\theta_r)}, \quad \theta_r \sim \mathbb{Q}.$$

- We can interpret the estimator as the expectation of g w.r.t. the random probability measure

$$\mathbb{P}^R(d\theta) = \sum_{r=1}^R W^{(r)} \delta_{\theta_r}(d\theta), \quad \theta_r \sim \mathbb{Q}.$$

- $\mathbb{P}^R$ is called the particle approximation of $\mathbb{P}$
 - $\{\theta_r, W^{(r)}\}_{r=1}^R$ is called weighted sample
- Weak convergence hold, for which $\mathbb{P}^R \Rightarrow \mathbb{P}$, i.e., for any bounded test function g ,

$$\mathbb{P}^R(g) = \sum_{r=1}^R W^{(r)} g(\theta_r) \xrightarrow{as} \mathbb{P}(g),$$

as $R \rightarrow \infty$. The rate of convergence is of order $O(1/\sqrt{R})$, as in MC approaches.

Measuring IS performance

- Similarly to what we have done with MCMC, we can assess the behavior of IS resorting to the effective sample size.

- For IS, **it can be expressed in a closed form** (Kong, 1992) as

$$\text{ESS}\left(\{W^{(r)}\}_{r=1}^R\right) = \frac{1}{\sum_{r=1}^R (W^{(r)})^2} = \frac{\left[\sum_{r=1}^R w(\theta_r)\right]^2}{\sum_{r=1}^R w(\theta_r)^2},$$

where $\text{ESS}(\{W^{(r)}\}_{r=1}^R) \in [1, R]$.

- $\text{ESS}\left(\{W^{(r)}\}_{r=1}^R\right) = R$ if $\mathbb{P} = \mathbb{Q}$, hence our proposal is the target density.
- The **closer is getting** $\text{ESS}\left(\{W^{(r)}\}_{r=1}^R\right)$ to R , **the better is our proposal** (and close to $\mathbb{P}$).
- In practice, once we collect our sample, we can assess its quality by computing empirically the $\text{ESS}(W_{1:R})$ with the observed weights.
- Further,

$$\lim_{R \rightarrow \infty} \text{ESS}/R = \frac{1}{1 + \chi^2(\mathbb{P} \parallel \mathbb{Q})}$$

with $\chi^2(\mathbb{P} \parallel \mathbb{Q}) = \int (\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{Q} - 1)^2 \mathrm{d}\mathbb{Q}$ being **the χ^2 divergence**.

- In MCMC, ESS accounts for autocorrelation between samples, in IS for weight variability among independent samples. Both measure information content.

Curse of dimensionality and IS

- It is a commonplace that importance sampling **suffers from the curse of dimensionality**.
- Consider again distributions $\mathbb{P}$ and $\mathbb{Q}$, with $\mathbb{P} \ll \mathbb{Q}$, and let

$$\mathbb{P}_p(d\theta_{1:p}) = \prod_{j=1}^p \mathbb{P}(d\theta_j), \quad \mathbb{Q}_p(d\theta_{1:p}) = \prod_{j=1}^p \mathbb{Q}(d\theta_j),$$

i.e. both $\mathbb{P}_p$ and $\mathbb{Q}_p$ **factorize in copies** of $\mathbb{P}$ and $\mathbb{Q}$, respectively.

- Hence, if we consider again $w(\theta) = d\mathbb{P}(\theta)/d\mathbb{Q}(\theta)$ (normalized here), we have

$$w_p(\theta_{1:p}) = \frac{d\mathbb{P}_p}{d\mathbb{Q}_p} = \prod_{j=1}^p w(\theta_j).$$

Then,

$$\text{Var}_{\mathbb{Q}_p}(w(\theta_{1:p})) = \mathbb{Q}(w^2)^p - 1,$$

which **grows exponentially in p number of dimensions**.

- As a consequence, the ESS degrades dramatically since

$$\text{ESS}/R = \frac{1}{R \sum_{r=1}^R (W^{(r)})^2} \approx \frac{1}{\mathbb{Q}(w^2)^p}.$$

The Pima Indian dataset

- We consider again the “famous” **Pima Indian** dataset, with $n = 532$ and $p = 8$.
- The purpose of this exercise is mainly to present the implementation of the various IS algorithms and show their performance in this specific example.
- **The following results should not be generalized** to any statistical models nor even to any logistic regression model.
- We consider once again a logistic model, with

$$y_i \mid \lambda_i \stackrel{ind}{\sim} \text{Bern}(\lambda_i), \quad \lambda_i = g(\eta_i), \quad \eta_i = \beta_1 x_{i1} + \cdots + \beta_p x_{ip},$$

and a priori an independent component Gaussian prior, with $\beta \sim N_p(\mathbf{0}, 100\mathbf{I}_p)$.

- We consider different proposal distributions.

Naive covariance matrix

- Let us start with a naive choice for the proposal covariance $S = 10^3 I_p$, centering the distribution in $\mathbf{0}$.
- This “random” choice of S works terribly, producing garbage results.

```
# Covariance matrix and expectation of the proposal
S <- diag(100, ncol(X))
mu <- rep(0, 8)

# Running the IS (R = 30000)
IS_not_informed <- IS(R, y, X, S, mu)

est_coef <- colSums(IS_not_informed$values * exp(IS_small$weights))
# X      Xnpreg  Xglu   Xbp  Xskin  Xbmi  Xped  Xage
# -0.866  0.433  1.201 -0.039  0.060  0.602  0.401  0.145

ESS <- 1 / sum(exp(IS_not_informed$weights)^2)
# 1

idx <- sample(1:R, R, TRUE, exp(IS_not_informed$weights))
length(unique(idx))
# 1
```

Informed covariance matrix

- We now set the proposal parameters according to the Laplace approximation of the posterior distribution.

```
# Covariance matrix and expectation of the proposal
S <- vcov(fit_logit)
mu <- coefficients(fit_logit)

# Running the IS (R = 30000)
IS_informed <- IS(R, y, X, S, mu)

est_coef <- colSums(IS_not_informed$values * exp(IS_small$weights))
# X      Xnpreg  Xglu   Xbp  Xskin  Xbmi  Xped  Xage
# -0.865  0.253  0.943 -0.068  0.234  0.424  0.324  0.397

ESS <- 1 / sum(exp(IS_not_informed$weights)^2)
# 2876.212

idx <- sample(1:R, R, TRUE, exp(IS_not_informed$weights))
length(unique(idx))
# 7378
```

Importance resampling

Motivation

- Resampling is the action of drawing randomly from a weighted sample, so as to obtain an unweighted sample.
- Resampling has the curious property of potentially reducing the variance at a later stage. This point is crucial for the good performance of particle algorithms.
- We have the following particle approximation of $\mathbb{P}_0(d\theta)$

$$\mathbb{P}_0^R(d\theta) = \sum_{r=1}^R W_0^{(r)} \delta_{\theta_0^{(r)}}, \quad \theta_0^{(r)} \sim \mathbb{Q}_0, \quad W_0^{(r)} = \frac{w_0(\theta_0^{(r)})}{\sum_{r=1}^R w_0(\theta_0^{(r)})},$$

obtained through importance sampling based on the proposal $\mathbb{Q}_0$ and the weight function w_0 .

- Ideally, we want to recycle the previous to approximate the following extended probability measure

$$(\mathbb{P}_0 Q_1)(d\theta_{0:1}) = \mathbb{P}_0(d\theta_0) Q_1(\theta_0, d\theta_1),$$

where $Q_1(\theta_0, d\theta_1)$ is a probability kernel function.

→ $Q_1(\theta_0, \cdot)$ is a probability measure

→ $Q_1(\theta_0, A)$ is measurable in θ_0

- There are two solutions to this problem.

Importance resampling I

- At first, we recognize that importance sampling from $\mathbb{Q}_1 = \mathbb{Q}_0 Q_1$ to $\mathbb{P}_0 Q_1$ requires
 - i) to sample $\{\theta_0^{(r)}, \theta_1^{(r)}\}_{r=1}^R$ from $\mathbb{Q}_0 Q_1$;
 - ii) to compute the weights, which are precisely $w_0(\theta_0^{(r)})$, $r = 1, \dots, R$, since $\theta_1^{(r)}$ conditionally on $\theta_0^{(r)}$ is sampled from the correct distribution.

- If we want to follow this strategy, the only thing is left to sample is

$$\theta_1^{(r)} \sim Q_1(\theta_0^{(r)}, d\theta_1), \quad r = 1, \dots, R.$$

- This strategy belongs to the **sequential importance sampling** framework.
- Inference relative to $(\mathbb{P}_0 Q_1)$ may be analyzed as standard importance sampling, without paying attention to the intermediate step.
- In practice, we are keeping fixed the weight values, but we resample our particles according to the probability kernel function Q_1 , filtering our particles over time.
- It is nothing but resampling with replacement from $\{\theta_0^{(r)}, W_0^{(r)}\}_{r=1}^R$, which is exactly the weighted bootstrap (Smith and Gelfand, 1992).

Importance resampling II

- The second strategy considers a two-step approximation. We first replace in the previous definition of the target extended probability measure $\mathbb{P}_0$ by $\mathbb{P}_0^R$, with

$$\mathbb{P}_0^R(d\theta_0)Q_1(\theta_0, d\theta_1) = \sum_{r=1}^R W_0^{(r)} Q_1(\theta_0^{(r)}, d\theta_1) \delta_{\theta_0^{(r)}}(d\theta_0).$$

- Then, we resample R times from the previous, which is an intermediate approximation, so we sample from

$$\frac{1}{R} \sum_{r=1}^R \delta_{\tilde{\theta}_{0:1}^{(r)}}, \quad \tilde{\theta}_{0:1}^{(r)} \sim \mathbb{P}_0^R(d\theta_0)Q_1(\theta_0, d\theta_1).$$

- This second approach is called **importance resampling**, as it is connected to other resampling techniques, such as the bootstrap, but is not anymore sampling with replacement.
- At first, it seems unappealing, as it has two approximation steps. Intuitively, it should have a larger Monte Carlo error and more computationally expensive. However, the first intuition turns out to be incorrect.

Toy example

- Let us consider the following toy example (Section 9.2, Chopin and Papaspiliopoulos, 2020). We set our sampling space $\Theta = \mathbb{R}$.

- Our approximating measure at time 0 is a standard Gaussian distribution, i.e. $\mathbb{Q}_0 \stackrel{d}{=} N(0, 1)$.
- Our target distribution is a truncated Gaussian distribution, setting mass in $[-\tau, +\tau]$, i.e. $\mathbb{P}_0 \stackrel{d}{=} TN_\tau(0, 1)$.
- $Q_1(\theta_0, d\theta_1)$ is our probability kernel function, with

$$\theta_1 = \rho\theta_0 + \sqrt{1 - \rho^2}U, \quad U \sim N(0, 1).$$

- We consider as test function $g(\theta_1) = \theta_1$, identity function, we want to obtain a sample at time 1.

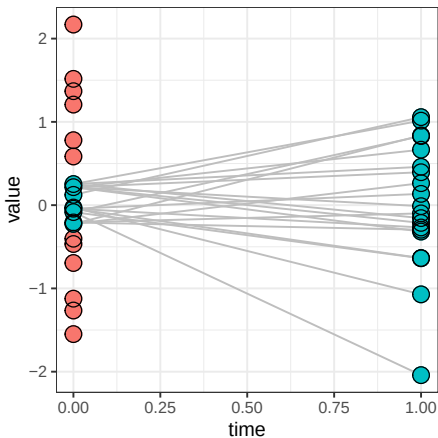
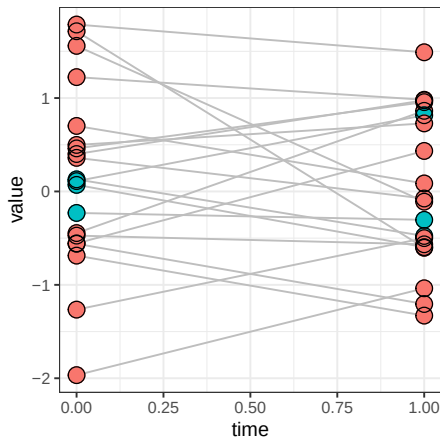
- We have two alternatives

importance sampling: $\hat{g}_{IS}(\theta) = \sum_{r=1}^R W_0^{(r)} \theta_1^{(r)}, \quad (\theta_0^{(r)}, \theta_1^{(r)}) \sim \mathbb{Q}_0 Q_1,$

importance resampling: $\hat{g}_{IR}(\theta) = \frac{1}{R} \sum_{r=1}^R \theta_1^{(r)}, \quad \theta_1^{(r)} \sim \mathbb{P}_1^R Q_1,$

where $\theta_1^{(r)} \sim \mathbb{P}_1^R Q_1$ means we draw from the corresponding marginal distribution.

Toy example



Effect of resampling for the toy example with $N = 20$, $\tau = 0.3$, $\rho = 0.5$. Green dots represent particles with non-zero weights.

Toy example

- For this specific case, we have that in the limit behavior

$$\sqrt{R}\hat{g}_{IS} \xrightarrow{d} N(0, \sigma_{IS}^2),$$

$$\sqrt{R}\hat{g}_{IR} \xrightarrow{d} N(0, \sigma_{IR}^2),$$

with

$$\sigma_{IS}^2 = \rho^2 \frac{\gamma(\tau)}{S(\tau)} + (1 - \rho^2) \frac{1}{S(\tau)},$$

$$\sigma_{IR}^2 = \rho^2 \frac{\gamma(\tau)}{S(\tau)} + (1 - \rho^2) + \rho^2 \gamma(\tau),$$

where $S(\tau) = \mathbb{Q}_0(w_0) = P(|\theta_0| \leq \tau) = 2\Phi(\tau) - 1$ and $\gamma(\tau) = \mathbb{P}_0(\theta_0^2)$. Note that

$$\rho^2 \gamma(\tau) < (1 - \rho^2) \left(\frac{1}{S(\tau)} - 1 \right),$$

hence, since as $\tau \rightarrow \infty$ we have $S(\tau) \rightarrow 0$ and the IR is better.

- The first term is due to the variability of the particles θ_0 at time 0, equal for both.
- The second term is due to the variability of the innovation terms U (simulated when sampling from kernel Q_1), smaller for importance resampling since $S(\tau) < 1$.
- The third term appears only for IR, and comes from the randomness introduced by the resampling step.
- Further, $(\sigma_{IS}^2 - \sigma_{IR}^2) \rightarrow +\infty$ for $\tau \rightarrow 0$ and any fixed ρ .

- We have a general structure to filter particles from time 0 to 1, which we can generalize to multiple times and combined with many possible model choices.
 - We can define general algorithms to deal with state-space models.
- Against our intuitions, resampling does not always result in an increase of the estimation variance. Indeed, sometimes, e.g. in presence of some hard constraints such as truncation, the variance can be reduced, as shown in the toy example.
- There are many possible sampling schemes, some of them can be better in specific scenarios. In practice, systematic sampling works pretty well.
- The resampling step introduces extra noise but eliminates the weight-variance problem. Whether resampling helps depends on the specific problem geometry
- Resampling can be viewed as random weights importance sampling.

Importance resampling with Pima dataset

- We consider again the Pima dataset, but now with the proposal distribution at time 0 being a multivariate Gaussian with parameter based on the Laplace approximation.
- Moving from time 0 to time 1, we resort to the transition kernel of the MALA algorithm.

```
# Covariance matrix and expectation of the proposal
S <- vcov(fit_logit)
mu <- coefficients(fit_logit)
sigma2 <- 0.1

# Running the IrS (R = 30000)
IrS_not_informed <- IrS(R, y, X, S, mu, sigma2)

est_coef <- colSums(IrS_not_informed$values * exp(IrS_small$weights))
# X      Xnpreg  Xglu   Xbp  Xskin  Xbmi  Xped  Xage
# -0.866  0.433  1.201 -0.039  0.060  0.602  0.401  0.145

ESS <- 1 / sum(exp(IrS_not_informed$weights)^2)
# 12362.98

idx <- sample(1:R, R, TRUE, exp(IrS_not_informed$weights))
length(unique(idx))
# 13740
```
